INCH-POUND
AA-C-101A
October 29, 2019
SUPERSEDING
AA-C-101
July 5, 2017

FEDERAL SPECIFICATION

CABINETS, HAZARDOUS MATERIAL (HM), SHIPBOARD QUALIFIED

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification establishes the requirements for hazardous material (HM) cabinets (formerly known as lockers) that are qualified for shipboard (both surface ship and submarine) use.

1.2 Classification. HM cabinets that are shock and vibration qualified are of the following types and classes, as specified (see 6.2 and 6.4):

1.2.1 Types.

a. Type I – Flammable cabinet, shock and vibration qualified, yellow
b. Type II – Corrosive cabinet, shock and vibration qualified, dark blue

1.2.2 Classes.

a. Class 1 – 12 gallons
b. Class 2 – 15 gallons
c. Class 3 – 30 gallons

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

MILITARY SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MILITARY STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 7125

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

FEDERAL REGULATIONS

Federal Acquisition Regulation (FAR)

(Copies of the FAR are available from the Superintendent of Documents, U.S. Government Publishing Office, Washington, DC 20402 or online at https://www.acquisition.gov/far.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30 - Flammable and Combustible Liquids Code

(Copies of this document are available from NFPA, 1 Batterymarch Park, Quincy, MA 02169-7471 or online at www.nfpa.org.)

3. REQUIREMENTS

3.1 Qualification. The shipboard qualified HM cabinets furnished under this specification shall be products qualified for listing on the applicable qualified products list (see 4.3 and 6.3).

3.2 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.3 Installation. Shipboard qualified HM cabinets shall be designed to be positively fixed to a deck or a foundation built on a ship's deck. The cabinet bottom pad or the cabinet mounting feet shall be designed to be welded to the ship's deck or welded to a foundation that is welded to the deck (see 4.5.13).

3.4 Design and construction. Shipboard qualified HM cabinets shall be designed to store groups of HM based on the primary hazard (such as flammables or corrosives). In general, design requirements specify features such as size, color, mounting methods, closing mechanism, locking ability, containment sumps, and shock and vibration qualification as specified in 3.5 and 3.6.

3.5 Physical requirements. Refer to table I for a summary of physical requirements.

3.5.1 Cabinet color. Color requirements of shipboard qualified HM cabinets shall be as follows (see 4.5.9):

a. Type I (all classes) - yellow, in accordance with industry standard

b. Type II (all classes) - dark blue, in accordance with industry standard

3.5.2 Identification plate(s)/label(s). Shipboard qualified HM cabinets shall be provided with a permanently attached plate(s) or label(s) affixed to the cabinet exterior containing the following information (see 4.5.10):

a. Manufacturer name

b. Type, class, and associated nomenclature of cabinet (i.e., type I, class 1, flammable cabinet, shock and vibration qualified)

c. Cabinet capacity, in gallons

d. Commercial and Government Entity (CAGE) number

e. Part or identification number (PIN) (see 6.4)

3.5.3 Hazard warning plate(s)/label(s). Shipboard qualified HM cabinets shall be provided with a permanently attached plate(s) or label(s) affixed to the cabinet exterior in accordance with this section (see 4.5.11). The label(s) or plate(s) shall be fabricated in accordance with the requirements of NFPA 30 and shall be provided with the following inscriptions and content identification label:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

a. For type I, flammable cabinet, shock and vibration qualified (all classes):

Inscription:
FLAMMABLE
KEEP FIRE AWAY

b. For type II, corrosive cabinet, shock and vibration qualified (all classes):

Inscription:
HAZARDOUS
KEEP FIRE AWAY

Content Identification Label:



CORROSIVE
STORAGE CABINET

3.5.4 Cabinet shelves. Shipboard qualified HM cabinets shall be provided with a minimum of two shelves per cabinet (see 4.5.8). Cabinet shelves shall be height adjustable.

3.5.5 Non-metallic materials of construction. All non-metallic materials of shipboard qualified HM cabinets shall be fully cured prior to government acceptance (see 4.5.12).

3.6 Performance requirements. Refer to table I for a summary of performance requirements.

3.6.1 Shock resistance. Type I and type II cabinets shall meet the grade B shock requirements of MIL-DTL-901 when tested in accordance with 4.5.1.

3.6.2 Vibration resistance. Type I and type II cabinets shall meet the type I vibration requirements of MIL-STD-167-1 when tested in accordance with 4.5.2.

3.6.3 Fire protection rating. Type I and type II cabinets shall meet the fire protection rating criteria specified in NFPA 30 when tested in accordance with 4.5.3.

3.6.4 Spill containment. Type I and type II cabinets shall have secondary containment or a cabinet sump that contains a minimum of 10 percent of the aggregate cabinet volume when tested in accordance with 4.5.4.

This spill containment requirement may be met by providing a 2-inch raised sill designed to retain liquid as long as the 2-inch sill depth equates to a sump design that can hold at least 10 percent of the aggregate cabinet volume. The 2-inch raised sill is required by NFPA 30 for flammable liquid storage cabinets.

3.6.5 Self-closing, self-latching doors. Type I and type II cabinets shall have self-closing, self-latching doors when tested in accordance with 4.5.5.

3.6.6 Lockability. Type I and type II cabinets shall be lockable by padlock or integral lock when tested in accordance with 4.5.6.

3.6.7 Steel with acid-resistant epoxy paint. Type II cabinets shall be of steel construction with acid-resistant epoxy paint or epoxy powder coating when tested in accordance with 4.5.7.

3.6.8 Lined cabinet shelves. Type II cabinet shelves shall be either galvanized metal or lined with acid-resistant material (epoxy paint or epoxy powder coating). In addition, type II cabinet shelves shall be fitted with polyethylene trays or liners to contain spillage when tested in accordance with 4.5.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall include the examinations and tests listed in table I. Any redesign or modification of the manufacturer's standard product not qualified in accordance with 4.3 shall constitute rejection of the cabinets.

4.4 Conformance inspection. Conformance inspection shall include the examinations and tests listed in table I.

4.4.1 Lot. All unmixed material of the same type offered for delivery at one time shall be considered a lot for purposes of conformance inspection.

4.4.2 Sampling for conformance testing. A sufficiently large production sample of each cabinet shall be selected in order to permit the performance of all conformance tests.

4.4.3 Small lots. Conformance tests shall not be required on any delivery of less than 100 cabinets. However, deliveries of such small quantities that have not been tested shall be identified in the manufacturer's records. Once the accumulated total of such untested deliveries reaches 100 cabinets, sample material shall be selected and subjected to the required conformance inspections identified in table I.

4.4.4 Inspection conditions. Specific inspection conditions shall not be required. Samples may be inspected under normal factory or warehouse conditions.

TABLE I. Qualification and conformance inspections.

Requirement	Requirement paragraph	Test method paragraph	Qualification inspection required	Conformance inspection required	Applicability (Type[s])
Shock resistance	3.6.1	4.5.1	X		I, II
Vibration resistance	3.6.2	4.5.2	X		I, II
Fire protection rating	3.6.3	4.5.3	X		I, II
Spill containment	3.6.4	4.5.4	X		I, II
Self-closing, self-latching doors	3.6.5	4.5.5	X	X	I, II
Lockable	3.6.6	4.5.6	X	X	I, II
Steel with acid-resistant epoxy paint	3.6.7	4.5.7	X		II
Minimum of two height adjustable shelves	3.5.4	4.5.8	X	X	I, II
Lined cabinet shelves	3.6.8	4.5.8	X	X	II
Color, type I	3.5.1	4.5.9	X	X	I
Color, type II	3.5.1	4.5.9	X	X	II

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

TABLE I. Qualification and conformance inspections – Continued.

Requirement	Requirement paragraph	Test method paragraph	Qualification inspection required	Conformance inspection required	Applicability (Type[s])
Identification plate(s)/labels(s)	3.5.2	4.5.10	X	X	I, II
Hazard warning plate(s)/label(s)	3.5.3	4.5.11	X	X	I, II
Non-metallic materials of construction	3.5.5	4.5.12	X		I, II
Installation	3.3	4.5.13	X		I, II

4.5 Test methods.

4.5.1 Shock. Type I and type II cabinets shall be tested to ensure they meet the grade B shock requirements of MIL-DTL-901. Test results shall be documented (see 6.2). Approval of test results shall be obtained from the Navy Preparing Activity and documented (see 6.2).

4.5.1.1 Test plan. A shock test plan shall be prepared. Approval of the shock test plan from the Navy Preparing Activity shall be obtained prior to initiating shock testing. This test plan may be combined with the vibration test plan (see 4.5.2.1).

4.5.1.2 Test type. Shock testing shall be for class 1, type A equipment in accordance with MIL-DTL-901. For purposes of the shock test, the following MIL-DTL-901 classifications shall be applied to type I and type II cabinets:

a. Test category – medium weight.

b. Mounting orientation – see table II.

c. Mounting requirements – An adapter plate or baseplate shall be used in this test. The baseplate shall be in accordance with MIL-DTL-901. The baseplate shall be a square plate at least 1 inch thick. The test fixture baseplate shall be attached to the standard fixtures on the medium weight shock test machine. The cabinet shall be welded to the baseplate.

TABLE II. Vertical axis specified test schedule.

Test axis	Group I	Group II	Group III
Vertical	X	X	X
30 degrees 1/		X	X
30 degrees, rotated 1/		X	X
NOTE: 1/ With cabinet door facing downwards toward floor.			

4.5.1.3 Loading of cabinet during test. During shock testing, the cabinet shall be loaded to its capacity with dummy weights. The dummy weights to be used in this testing are 1-gallon cans filled with water. The cans shall be filled using a graduated measuring cup or beaker so that the liquid level is as close to 1 gallon as possible. The water in the containers shall be dyed to make it more visible to observers and in photographs. The cans shall be evenly distributed on the shelves, but cans shall not be double stacked on the same shelf. The containers to be placed inside the cabinet during testing shall be a combination of new, unused metal paint cans with lids and metal jugs with caps. Each paint can shall have the lid secured using six lid clips per gallon can. After loading the cabinet, the door(s) of the cabinet shall be closed and latched, but not padlocked or secured by any other external means.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

4.5.1.4 Monitoring during shock test. The cabinet being tested shall be monitored visually after each hammer blow. Damage shall be noted by the test technician in writing in the test log and with photographs. Neither instrumentation nor transducers will be required during the testing. The cabinet door shall not be opened at any point during the seven test blows to inspect the dummy weights inside the cabinet.

4.5.1.5 Post-shock test inspection.

a. At the completion of the test blows, the baseplate and the cabinet shall be removed from the test fixture and placed on a level surface. At this point, the cabinet door shall be opened, and the contents of the cabinet shall be documented with photographs. The dummy weights shall be removed from the cabinet; use care when removing the dummy weights so as not to spill additional liquid. All the liquid remaining in the dummy weight cans shall be measured and recorded. In addition, any liquid in the cabinet sump shall be vacuumed out and its volume measured and recorded.

b. When the cabinet is empty of the dummy weights and any remaining liquid has been vacuumed out, the interior of the cabinet shall be visually inspected. Damage shall be recorded in writing in the test log and with photographs.

c. A leak test of the cabinet sump shall be conducted. The sump shall be filled to its capacity with liquid and observed for at least 15 minutes to see if the sump is leaking.

4.5.2 Vibration. Type I and type II cabinets shall be tested to ensure they meet the type I requirements of MIL-STD-167-1 for environmental vibration within a test range of 4 to 50 Hertz (Hz). Test results shall be documented (see 6.2). Approval of the test results shall be obtained from the Navy Preparing Activity and documented (see 6.2).

4.5.2.1 Test plan. A vibration test plan shall be prepared. Approval of the vibration test plan from the Navy Preparing Activity shall be obtained prior to initiating vibration testing. This test plan may be combined with the shock test plan (see 4.5.1.1).

4.5.2.2 Test type. For purposes of the vibration test, the following shall be applied to type I and type II cabinets:

a. Mounting requirements – An adapter plate or baseplate shall be used in this test. The baseplate shall be a square plate at least 1 inch thick. The test fixture shall be attached to the platform of the vibration test machine. The cabinet shall be welded to the baseplate.

b. Equipment orientation – For these tests, the vibration axes shall be vertical, horizontal front-to-back, and horizontal side-to-side.

c. Exploratory, variable frequency, and endurance testing shall be conducted.

4.5.2.3 Loading of cabinet during test. During vibration testing, the cabinet shall be loaded to its capacity with dummy weights. The dummy weights to be used in this testing are 1-gallon cans filled with water. The cans shall be filled using a graduated measuring cup or beaker so that the liquid level is as close to 1 gallon as possible. The water in the containers shall be dyed to make it more visible to observers and in photographs. The cans shall be evenly distributed on the shelves, but cans shall not be double stacked on the same shelf. The containers to be placed inside the cabinet during testing shall be a combination of new, unused metal paint cans with lids and metal jugs with caps. Each paint can shall have the lid secured using six lid clips per gallon can. After loading the cabinet, the door(s) of the cabinet shall be closed and latched, but not padlocked or secured by any other external means.

4.5.2.4 Monitoring during vibration test. The cabinet being tested shall be monitored visually and with accelerometers during the exploratory and variable frequency tests to determine if resonances exist.

4.5.2.5 Post-vibration test inspection.

a. At the completion of all testing in all axes, the cabinet door shall be opened, and the contents of the cabinet shall be documented with photographs. The dummy weights shall be removed from the cabinet; use care when removing the dummy weights so as not to spill additional liquid. All the liquid remaining in the dummy weight cans shall be measured and recorded. In addition, any liquid in the cabinet sump shall be vacuumed out and its volume measured and recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

b. When the cabinet is empty of the dummy weights and any remaining liquid has been vacuumed out, the interior of the cabinet shall be visually inspected. Damage shall be recorded in writing in the test log and with photographs.

c. A leak test of the cabinet sump shall be conducted. The sump shall be filled to its capacity with liquid and observed for at least 15 minutes to see if the sump is leaking.

4.5.2.6 Acceptance criteria.

a. The cabinet door shall not open, and the cabinet structure shall remain intact. All internal and external parts and accessories shall remain attached. The cabinet door shall be able to be opened at the conclusion of the testing.

b. External spillage of contents shall not be permitted. Ruptured containers inside the cabinet shall not by themselves constitute a failure if the cabinet sump has retained the spill liquid.

c. Sump joints shall maintain their integrity and be capable of retaining liquids.

4.5.3 Fire protection rating. Evidence that the cabinet meets the criteria specified in NFPA 30 shall be as specified (see 6.2).

4.5.4 Spill containment. Evidence that the containment or sump holds a minimum of 10 percent of the aggregate cabinet volume for type I and type II cabinets shall be as specified (see 6.2). In addition, a sump leak test shall be conducted and test results documented. For the leak test, the cabinet shall be placed on a level surface. The sump shall be filled until it is on the verge of overflowing. The volume (in gallons) of water in the sump shall be recorded. With the cabinet sump filled to the verge of overflow, the cabinet shall stand for 15 minutes with no signs of external leakage.

4.5.5 Self-closing, self-latching doors. A functional test shall be conducted and test results documented (see 6.2) as evidence that cabinets have self-closing, self-latching doors.

4.5.6 Lockable. A functional test shall be conducted and test results documented (see 6.2) as evidence that cabinet doors can be locked.

4.5.7 Steel with acid-resistant epoxy paint. Evidence that cabinets are steel with acid-resistant epoxy paint shall be as specified (see 6.2).

4.5.8 Cabinet shelves. Evidence that cabinets are supplied with a minimum of two shelves shall be as specified (see 6.2). Documentation (see 6.2) shall also provide evidence that cabinet shelves are either galvanized metal or lined with acid-resistant epoxy paint or epoxy powder coating. The documentation shall include evidence that the cabinet shelves are fitted with polyethylene trays or liners to contain spillage. In addition, direct observation of the cabinet shelves shall be allowed.

4.5.9 Color. Evidence that cabinets are in compliance with the color requirements (see 3.5.1) shall be as specified (see 6.2). In addition, direct observation of cabinet color shall be allowed.

4.5.10 Identification plate(s)/label(s). Evidence that cabinets have an identification plate(s)/label(s) (see 3.5.2) shall be as specified (see 6.2). In addition, direct observation of the cabinet identification plate(s)/label(s) shall be allowed.

4.5.11 Hazard warning plate(s)/label(s). Evidence that cabinets have a hazard warning plate(s)/label(s) (see 3.5.3) shall be as specified (see 6.2). In addition, direct observation of the cabinet hazard warning plate(s)/label(s) shall be allowed.

4.5.12 Non-metallic materials of construction. Evidence that cabinet materials (non-metallic) are fully cured (see 3.5.5) prior to government acceptance shall be as specified (see 6.2).

4.5.13. Installation. Evidence that cabinets are designed to be installed as specified in 3.3 shall be as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2).

6. NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful but is not mandatory.)

6.1 Intended use.

a. Type I, class 1, class 2, and class 3 flammable cabinets are intended for storage of a limited supply of flammable and/or combustible material onboard ship. These cabinets are shock and vibration qualified. Limitations regarding shipboard installation locations can be found in Naval Ships' Technical Manual S9086-WK-STM-020, NSTM Chapter 670, Stowage, Handling, and Disposal of Hazardous General Use Consumables, Volume 2 - Afloat Hazardous Material Control and Management Guidelines Hazardous Materials Users Guide (HMUG).

b. Type II, class 1, class 2, and class 3 corrosive cabinets are intended for storage of a limited supply of corrosive material onboard ship. These cabinets are shock and vibration qualified. Limitations regarding shipboard installation locations can be found in S9086-WK-STM-020/670.

6.2 Acquisition requirements.

a. Title, number, and date of this specification.

b. Type and class required (see 1.2).

c. The specific issue of individual documents referenced (see 2.1 and 2.2).

d. Documentation (drawings, reports, etc.) to be submitted as evidence that requirements have been met (see 4.5.1 through 4.5.13).

e. Packaging requirements (see 5.1).

f. Requirements for installation instructions (see 6.5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the supplier is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or CommandStandards@navy.mil, and information pertaining to qualification of products may be obtained from that activity.

6.4 PIN. The following PIN procedure is for government purposes and does not constitute a requirement for the contractor:

F	**AA**	**-**	**C**	**-**	**101**	**-**	**X**	**X**
Prefix for Federal Specification	Alphabetical designation	-	Alphabetical designation	-	Specification number	-	Type (see 6.4.1)	Class (see 6.4.2)
Examples: The PIN for a flammable shipboard qualified HM cabinet with a maximum capacity of 12 gallons would be FAA-C-101-A1. The PIN for a corrosive shipboard qualified HM cabinet with a maximum capacity of 15 gallons would be FAA-C-101-B2.								

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.

6.4.1 Types. The cabinet types PIN codes are as follows:

Type	Characteristic	Color	PIN Code
Type I	Flammable cabinet, shock and vibration qualified	Yellow	A
Type II	Corrosive cabinet, shock and vibration qualified	Dark blue	B

6.4.2 Classes. The cabinet classes PIN codes are as follows:

Class	Characteristic	PIN Code
Class 1	12 gallons	1
Class 2	15 gallons	2
Class 3	30 gallons	3

6.5 Drawings and instructions. Drawings, sketches, and instructions should provide installation foundation, mounting, and applicable corrosion control details (see 6.2).

6.6 Subject term (key word) listing.

Acid
Combustible
Corrosive
Flammable
Locker
Shock
Vibration

6.7 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

MILITARY INTERESTS

Custodians:
Army – MI
Navy – SH
Air Force – 03

Review Activities:
Army – AV, GL, QM1
Navy – CG, YD

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy − SH
(Project 7125-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:38Z
Check the source to verify that this is the current version before use.